

Noise Computation

Costanzo2016 Fitness and Variance Computation

Critical Quotes from SI

Line 74 (Double mutant):

“All screens were conducted a single time with **4 replicate colonies per double mutant**, unless otherwise indicated”

Line 88 (Single mutant):

“Colony size measurements of SGA deletion and TS array mutant strains were based on an average of **350 replicate control screens**... Colony size measurements of SGA deletion and TS query mutant strains were based on an average of **17 replicate control screens**... Colony size measurements were used to estimate single mutant fitness as described previously (5) with the exception that **bootstrapped means, instead of medians, across replicates were used in variance estimation and final fitness values.**”

Key: “Replicates” for SMF = **screens** (17 or 350), NOT individual colonies!

Measurement Summary

Measurement	n_screens	n_colonies/screen	Total measurements	Variance method	Reported “std”	n_replicates
Query SMF	17	4	68	Bootstrap across screens	SE	17
Array SMF	350	4	1400	Bootstrap across screens	SE	350
DMF	1	4	4	Raw sample SD	Raw SD	4

Single Mutant Fitness (SMF) - Bootstrap Procedure

Hierarchical structure:

Screen 1: [col1, col2, col3, col4] → mean → screen1_value

Screen 2: [col1, col2, col3, col4] → mean → screen2_value

...

Screen N: [col1, col2, col3, col4] → mean → screenN_value

(N = 17 for query, N = 350 for array)

Bootstrap: Resample with replacement from [screen1, ..., screenN]

→ Compute mean of resampled screens

→ Repeat B times (e.g., B=1000)

→ Report: mean of bootstrap means, SD of bootstrap means

Key insight: Bootstrap SD is already SE-like because it's the SD of the **sampling distribution of the mean**.

$$\sigma_{bootstrap}(f) \approx SE(f) = \frac{\sigma_{among_screens}}{\sqrt{N}}$$

where $N = 17$ (query) or 350 (array).

Double Mutant Fitness (DMF) - Raw Sample Statistics

Computation (not explicitly stated, but likely):

$$\bar{f}_{ij} = \frac{1}{4} \sum_{k=1}^4 f_{ij,k}$$
$$\sigma(f_{ij}) = \sqrt{\frac{1}{3} \sum_{k=1}^4 (f_{ij,k} - \bar{f}_{ij})^2}$$

No bootstrap - just raw sample SD across 4 colonies from one screen.

To get SE:

$$SE(f_{ij}) = \frac{\sigma(f_{ij})}{\sqrt{4}} = \frac{\sigma(f_{ij})}{2}$$

Why n_replicates = 17/350, NOT 68/1400?

Wrong interpretation: Total measurements = $17 \times 4 = 68$ or $350 \times 4 = 1400$

Correct interpretation:

- Bootstrap resamples **screens as units**, not individual colonies
- 4 colonies per screen are **averaged before** bootstrap resampling
- Colonies on same plate = technical replicates (same environment, same day)
- Screens on different days = biological replicates (capture day-to-day variation)
- **Bootstrap needs biological variation** → resamples screens, not colonies

Therefore: n_replicates = number of screens (17 or 350)

Implications for Error Propagation

For gene interactions: $\varepsilon = f_{ij} - f_i \cdot f_j$

$$Var(\varepsilon) = Var(f_{ij}) + f_j^2 Var(f_i) + f_i^2 Var(f_j)$$

The mixing problem:

Term	Variance	Interpretation
$Var(f_i)$	σ_i^2 (bootstrap)	Already SE^2 (accounts for $n=17$)
$Var(f_j)$	σ_j^2 (bootstrap)	Already SE^2 (accounts for $n=350$)
$Var(f_{ij})$	σ_{ij}^2 (raw SD)	Must divide by $n=4$ to get SE^2

Why $SE = \sigma/\sqrt{n}$ fails:

- For SMF: Over-corrects (bootstrap SD is already SE)
- For DMF: Correct (raw SD needs division by $\sqrt{n}$)

This explains why we couldn't reproduce p-values - we're mixing two different variance estimation procedures!

Schema Implementation

Store `n_replicates` as:

- Query SMF: `n_replicates = 17` (screens)
- Array SMF: `n_replicates = 350` (screens)
- DMF: `n_replicates = 4` (colonies)

Document clearly:

```
n_replicates: int | None = Field(
    default=None,
    description="""Number of independent units used to compute mean.

    WARNING: Interpretation of fitness_std differs by measurement type:
    - Costanzo SMF: Bootstrap SD (already SE-like, n=screens)
    - Costanzo DMF: Raw sample SD (divide by sqrt(n) for SE, n=colonies)

    See: torchcell.datasets.scerevisiae.costanzo2016.noise-computation
    """)
)
```

References

- SI Line 74: Double mutant screening (4 colonies per screen)
- SI Line 88: Single mutant fitness bootstrap (17 or 350 screens)
- SI Lines 572-575: Data file descriptions